

AI/ML intern DAY 19

AI/ML Internship Suggestions for Raritone

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Introduction

The AI/ML Internship involved researching and understanding several advanced technologies related to Artificial Intelligence, Computer Vision, Virtual Try-On systems, fashion datasets, body measurement extraction, and recommendation systems.

Throughout the internship, multiple technical and research challenges were encountered while studying complex AI models and understanding how different modules interact within a complete fashion technology platform.

Rather than viewing these difficulties as obstacles, they became valuable learning opportunities that improved my technical knowledge, analytical thinking, research methodology, and problem-solving abilities.

1. Challenge: Understanding the Complete Virtual Try-On Pipeline

Problem

At the beginning of the internship, it was difficult to understand how multiple technologies such as body detection, segmentation, pose estimation, garment warping, image blending, and recommendation systems work together to produce a realistic Virtual Try-On experience.

The overall workflow appeared highly complex because each module depends on the successful output of the previous stage.

Solution

To overcome this challenge, I studied the Virtual Try-On pipeline module by module instead of trying to understand the entire system at once.

I analysed workflow diagrams, compared different AI architectures, and documented the function of every stage individually.

Outcome

This approach helped me build a clear understanding of the complete AI workflow used in modern Virtual Try-On platforms.

2. Challenge: Learning Advanced Computer Vision Models

Problem

Several AI models introduced during the internship involved advanced Computer Vision concepts that were unfamiliar initially.

Some of these included:

- MediaPipe Pose
- Detectron2
- DeepLabV3
- SCHP
- OpenPose

Understanding their architectures, strengths, and practical applications required significant research effort.

Solution

I studied official documentation, research papers, model comparison articles, and workflow diagrams to understand the purpose of each model.

Instead of focusing only on implementation, I concentrated on understanding the role of every model within the Virtual Try-On ecosystem.

Outcome

I developed a strong conceptual understanding of modern Computer Vision systems and their practical applications.

3. Challenge: Body Landmark Detection and Measurement Extraction

Problem

Estimating body dimensions from 2D images was one of the most challenging topics during the internship.

Understanding how AI models calculate measurements such as:

- Height
- Shoulder Width
- Waist Width
- Hip Width
- Arm Length

required learning body landmark detection principles and geometric calculations.

Solution

I researched MediaPipe Pose landmarks and studied how distances between body keypoints can be used for measurement estimation.

Breaking the workflow into smaller stages made the concepts easier to understand.

Outcome

I gained a clear understanding of how body measurements can improve garment fitting and personalized recommendations.

4. Challenge: Studying Multiple Fashion Datasets

Problem

The internship required analysing several large-scale public datasets including:

- DeepFashion
- DeepFashion2
- VITON
- VITON-HD
- Dress Code
- MPV Dataset
- Fashionpedia

Each dataset had different structures, annotations, and use cases, making comparison difficult.

Solution

I created comparison tables and organized the datasets according to their primary applications such as segmentation, landmark detection, or Virtual Try-On.

I also studied metadata generation and annotation techniques to better understand dataset preparation.

Outcome

This systematic approach helped me understand the importance of high-quality datasets in AI model development.

5. Challenge: Understanding Garment Warping Techniques

Problem

Garment warping involves complex image transformation methods that initially appeared difficult to understand.

Advanced techniques such as:

- Thin Plate Spline (TPS)
- Geometric Matching Module (GMM)
- Flow-Based Warping
- Affine Transformation

require knowledge of both mathematics and Computer Vision.

Solution

Rather than focusing on mathematical details alone, I studied the practical purpose of each technique and compared how different Virtual Try-On models use them.

Visual workflow diagrams and architecture comparisons helped simplify these concepts.

Outcome

I gained a practical understanding of how AI systems adapt clothing to different body shapes while preserving garment appearance.

6. Challenge: Understanding 3D Avatar Generation

Problem

Technologies such as SMPL, PIFuHD, ICON, HumanNeRF, and 3D Gaussian Splatting involve advanced 3D reconstruction concepts that were completely new to me.

Understanding mesh generation, neural rendering, and human body reconstruction required extensive research.

Solution

I studied each model individually and compared their architectures, applications, and limitations instead of attempting to understand all of them simultaneously.

Outcome

This approach significantly improved my understanding of AI-driven 3D avatar generation and its future applications in digital fashion.

7. Challenge: Hardware and Resource Limitations

Problem

Many modern AI frameworks require powerful GPUs, large datasets, and high computational resources that were not always available during the internship.

Practical implementation of advanced models such as HR-VITON and PIFuHD would require significant hardware capabilities.

Solution

I shifted my focus toward research, comparative analysis, architecture studies, and workflow understanding while using lightweight implementations and demonstrations wherever possible.

Outcome

This allowed me to gain substantial knowledge about advanced AI technologies despite hardware limitations.

8. Challenge: Managing Multiple Research Topics

Problem

The internship covered many different domains simultaneously, including:

- Computer Vision
- Virtual Try-On
- Recommendation Systems
- Fashion Datasets
- 3D Reconstruction

- Garment Warping
- AI Documentation

Managing multiple topics while maintaining organized documentation became challenging.

Solution

I divided the internship into smaller research modules and maintained structured notes, reports, and GitHub repositories for each completed task.

Outcome

This improved my time management skills and helped me complete all assigned activities in an organized manner.

9. Challenge: Technical Documentation and GitHub Management

Problem

Preparing professional documentation and maintaining a properly structured GitHub repository required attention to detail and consistency.

Ensuring that reports, README files, screenshots, and implementations were properly organized was an important challenge.

Solution

A standardized documentation format was followed throughout the internship. Every module included objectives, research findings, conclusions, and supporting materials.

GitHub repositories were organized using day-wise folders and descriptive README files.

Outcome

This improved my technical writing abilities and introduced me to professional project documentation practices.

10. Key Strategies Used Throughout the Internship

The following approaches were consistently used to overcome challenges:

- Independent learning.
- Research paper analysis.

- Comparative study of AI models.
 - Workflow-based understanding.
 - Structured technical documentation.
 - Modular learning approach.
 - Continuous review and revision.
 - GitHub project organization.
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11. Lessons Learned

The challenges encountered during the internship became valuable learning experiences that strengthened both technical and professional skills.

I learned that solving complex AI problems requires patience, continuous learning, structured research, and the ability to break large problems into smaller manageable tasks.

The internship also demonstrated that understanding concepts deeply is often more important than simply implementing code.

12. Conclusion

Throughout the AI/ML Internship, several technical and research challenges were encountered while studying modern Artificial Intelligence applications in the fashion industry. These challenges included understanding Computer Vision models, body landmark detection, fashion datasets, garment warping techniques, recommendation systems, and 3D avatar generation.

By applying systematic research methods, comparative analysis, and structured documentation practices, each challenge was successfully transformed into a valuable learning opportunity.

Overall, the experience strengthened my problem-solving abilities, technical research skills, documentation practices, and confidence in exploring emerging AI technologies, providing a solid foundation for future academic and professional growth in Artificial Intelligence and Computer Vision.